Return to Full Participation Following Concussion in the National Football League, 2015 Through 2020

Christina D. Mack, PhD, MSPH,* Mackenzie M. Herzog, PhD, MPH,* Gary Solomon, PhD,†‡ Margot Putukian, MD, FACSM, FAMSSM,§ Rebecca Y. Lee, PhD,* Matthew J. Matava, MD,¶ Javier Cárdenas, MD,‖ Nicholas Theodore, MD,** and Allen Sills, MD‡

Abstract

Objective: Quantify days missed, games missed, injury burden, and time to return to full participation (RTFP) among National Football League (NFL) players who sustained a concussion. **Design:** Retrospective cohort study. **Setting:** 2015 through 2020 NFL seasons. **Participants:** National Football League players diagnosed with a concussion from 2015 to 2020. **Interventions:** National Football League-mandated graduated RTFP protocol. **Main Outcome Measures:** Days missed, games missed, burden, and time to RTFP, overall and by position. **Results:** An annual average of 3639 player-days of participation and 255 games were missed across NFL because of concussion. Concussed players missed a median of 9 days (mean = 15.0), a relatively stable metric over 6 years, with slight variation by position. Offensive linemen, tight ends, running backs, and linebackers missed the most days per concussion; defensive secondary, offensive linemen, and wide receivers sustained the highest injury burden. Post-concussion, 59% of players missed one or more scheduled games. Among players concussed in a Sunday game, 38% played in a Sunday game one week later. **Conclusions:** The 9-day median time missed post-concussion may be related to emphasis on graduated phase-based concussion management. No concussed player returned to competition on the day of injury, and less than 40% participated in games the following week. Further work is needed to better understand characteristics of concussions that take longer to return and movement through stages of return.

Key Words: American football, National Football League, concussion, epidemiology, return to play

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INTRODUCTION

Expert opinions about return to full participation (RTFP) after sport-related concussion (SRC) have evolved over 80 years, from individual recommendations by sports medicine physicians[1–4] to formal guidelines in the 1980s.[5] As guidelines and statements have progressed, attention to when and how patient-athletes return to sport have garnered increased scrutiny.

Consensus guidelines first appeared in the 1990s.[6–8] The first International Conference on Concussion in Sport (CISG) meeting was held in 2001[9]; a graduated, stepwise RTFP protocol was proffered, rooted in previously published Canadian Academy of Sport Medicine guidelines.[10] This protocol articulated the sequential steps clinically required for return, steering overall time to return. In 2005, the first Team Physician Consensus Statement addressing concussion was published,[11] representing the approach to concussion in the United States.

National Football League (NFL) RTFP protocols were first published by the NFL Head, Neck, and Spine (HNS) Committee in 2013[12,13] and updated in 2018.[14] Return to full participation practices are reviewed annually, with continued protocol revisions to align NFL operations with CISG statements, updated recommendations, and clinical precautions. This paper presents League-wide data on time to RTFP for concussed NFL athletes from 2015 through 2020 by assessing days and games missed among players and across the League.

METHODS

Data Source

This retrospective, observational study includes team-related concussions sustained by NFL players during the 2015

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*From the *Real World Solutions, IQVIA, Research Triangle Park, North Carolina; †Department of Neurological Surgery and Vanderbilt Sports Concussion Center Vanderbilt University School of Medicine, Nashville, Tennessee; ‡National Football League Player Health and Safety Department, New York, New York; §Massachusetts General Hospital, Harvard University, Boston, Massachusetts; ¶Department of Orthopedic Surgery, Washington University, St. Louis, Missouri; ‖Department of Neurology, Barrow Neurological Institute, Phoenix, Arizona; and **Department of Neurological Surgery, Johns Hopkins University School of Medicine, Baltimore, Maryland.*

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Corresponding Author: Christina D. Mack, PhD, MSPH, FISPE, Real World Solutions, IQVIA, 4820 Emperor Blvd, Durham, NC 27703 (Christina.Mack@IQVIA.com).

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through 2020 seasons. Concussions were diagnosed according to NFL protocols[11,12] Data were collected from the NFL's electronic health record (EHR), which include rigorously-audited injury information entered by Club medical staff (team physicians and certified athletic trainers).[15,16] EHR data were linked to NFL's Game Statistics and Information System participation data, which provide information on team-games that occurred while players were unavailable to participate and player employment with their Clubs. Study approval was obtained from [Mount Sinai] Institutional Review Board (#0006018123) and the NFL–NFL Players Association (NFLPA) Player Scientific and Medical Research Approval Process.[17]

Days Missed

Return to full participation is defined as the ability to participate in competition and all practice environments, including contact practices. Return to full participation is recorded regardless of timing in season and is not dependent on return to competition (ie, players can be designated as able to participate in full football activities even if a competition is not scheduled or they are not placed on a gameday roster). Days missed were calculated as the number of days that elapsed from the initial date of removal from participation because of the concussion diagnosis to the date the player returned to team activities. The return to team activities determination was based on the NFL protocol and involved 3 sequential factors: (1) the player successfully completing all phases of the RTFP protocol, (2) medical clearance from the head team physician, and (3) medical clearance from the independent neurologic consultant.[14] Concussions among active players that occurred in 2020 and were still out because of injury at season end (n = 8) were assumed to have missed all remaining days through the 2020 season end (February 12, 2021). Other injuries missing an RTFP date (n = 118; 8%) were excluded from days-missed calculations.

Games Missed

Number of games missed was defined as the number of games the player's team played between participation removal date to first game after RTFP (exclusive). For game concussions, the game in which the concussion occurred is not considered a missed game. Similar to the days missed methodology, concussions among players that had not returned at study end (n = 8) were assumed to miss all remaining games in their Club's 2020 season. Other injuries missing an RTFP date or no game played after RTFP (n = 185; 13%) were excluded from games-missed calculations. Number and percentage of players returning to their team's scheduled games by day of week and number of days missed was computed for regular season concussions.

Injury Burden

Two burden metrics were calculated to estimate aggregate impact of concussions across the league in lost participation based on total number of diagnosed concussions and sum of time lost: (1) total days missed (including offseason days) across all concussed players, inclusive of initial and additional days missed because of the concussion if a player was subsequently removed after initial return and (2) total games missed across all concussed players. In cases of missing RTFP information (days: n = 118; 8%; games: n = 185, 13%), missed time was imputed by median across the specific population of interest. Imputation by mean was computed as a sensitivity analysis; only numbers imputed by median are presented as results remained similar (3.4% increase in days missed, 0.5% increase in games missed).

Return to Full Participation Protocol

Figure 1 portrays the Return to Participation protocol, which was adapted from the CISG model[18–21] and approved by the NFLPA and NFL HNS Committee.[12]

Statistical Analysis

Descriptive statistics [mean, SD, median, interquartile range (IQR)] were calculated for days and games missed and stratified by year, competitive season (offseason, preseason, regular season, postseason), team activity (game, practice, other), roster position (offensive line, running back, tight end, quarterback, wide receiver, defensive secondary, defensive line, linebacker, special teams), position at time of concussion, and day of week (separately for game and practice). Medians are reported as the primary metric, because time missed does not follow a normal distribution. Wilcoxon rank-sum tests and Kruskal–Wallis tests were performed to compare days and games missed by characteristics of interest, such as practice/game and day of week. Results were considered statistically significant at $P < 0.05$.

RESULTS

From 2015 to 2020, 1480 concussions occurred to 1121 players during NFL team-related activities, ranging from 223 to 302 per year 2015 to 2019, and 178 in 2020 (shortened season without offseason practices/preseason games because of COVID-19) (Table 1).[22]

Days Missed

The distribution of days missed is shown in Figure 2. Players missed a median of 9 days (mean: 15.0; SD = 23.7) because of concussion from 2015 to 2020, with an average of 3639 days missed per year (range: 2180-5029; Table 2). 2017 was the highest burden year and 2020 was the lowest. Eleven concussions (<1%) required a second removal after initial RTFP because of new symptoms, missing additional time (median = 9, IQR 6, 18 days), without a new concussion diagnosis, 3 days (median) after RTFP (IQR 2, 5 days).

Median days missed was similar when comparing practice concussions (8 days) to game concussions (9 days, $P = 0.40$). The day that the injury occurred, which is a proxy for game schedule as a majority of games take place on Sundays, did not affect median days missed for games (Saturday-Tuesday = 9 days, Wednesday-Friday = 9 days, $P = 0.13$) or practices (Sunday-Wednesday = 8 days, Thursday-Saturday = 7.5 days, $P = 0.51$). Median days missed was similar across the NFL season (offseason = 8.5 days, preseason = 8 days, regular season = 9 days, postseason = 9 days; Table 2) and over the study period [8 days for 2018, 9 days for all other years (Figure 3)].

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Figure 1. National Football League Return to Full Participation protocol (July 2020 version).

TABLE 1. Distribution of Concussions in the NFL by Season and Session, 2015 Through 2020

	Number of Concussed Players	Number of Concussions										
			Preseason			Regular Season			Postseason			Total
	Total	Offseason	Game	Practice	Other	Game	Practice	Other	Game	Practice	Other	
2015	263	5	54	29	0	183	9	0	4	0	0	284
2016	242	5	45	26	0	166	6	0	6	1	0	255
2017	277	10	46	45	1	178	12	0	10	0	0	302
2018	206	5	34	45	1	127	8	0	3	0	0	223
2019	226	8	49	30	0	136	9	0	5	1	0	238
2020*	169	—	—	30	0	129	13	0	6	0	0	178
Total*	1121†	33	228	205	2	919	57	0	34	2	0	1480

** Due to the COVID-19 pandemic, there was no offseason or preseason games in 2020.*
† The total number of concussed players is less than the sum of players concussed each year because a player can sustain concussions in multiple years.

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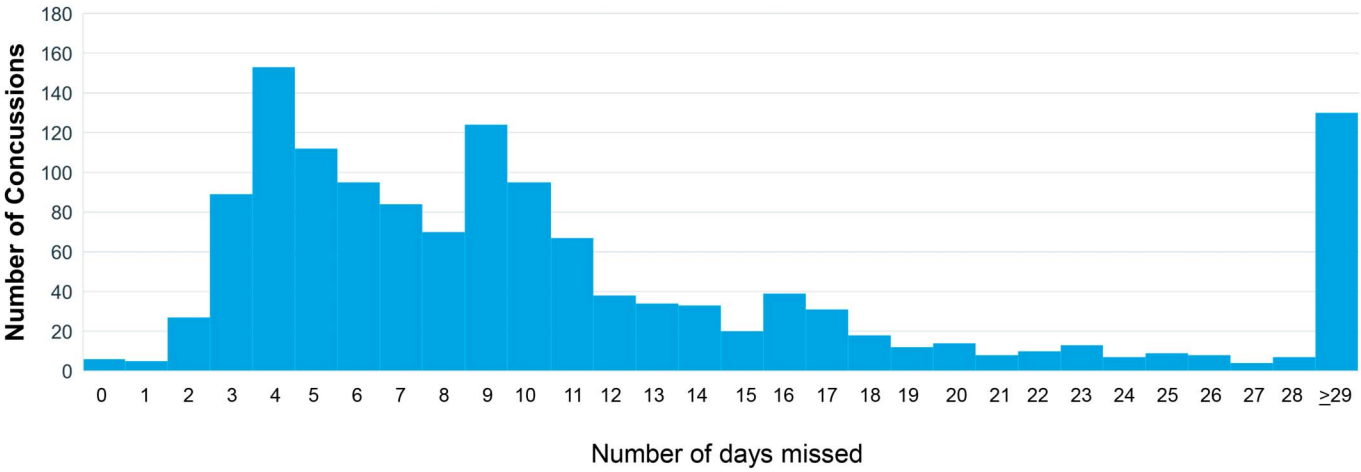

Figure 2. Distribution of days missed because of concussions in the National Football League, 2015 through 2020. Players sustaining a concussion in 2020, but had not yet returned to football activity at the time of manuscript submission and did not leave the club are assumed to have a return date of February 12, 2021 (n = 8). Players with concussions missing a RTFP were excluded from calculations of median days missed (n = 118).

There was a statistically significant difference in days missed by roster position ($P < 0.01$). Offensive linemen, tight ends, running backs, and linebackers had the highest median days missed per concussion (10, 9.5, 9, and 9 days; Table 2). Quarterbacks and special teams players missed the fewest days (7, 4 days). Defensive secondary players, offensive linemen, and wide receivers had the highest annual burden (761, 722, and 561 days, based on 366, 261, and 222 concussions). By position at time of injury, players participating on special teams plays missed a median of 9 days with a burden of 492 days missed per year (225 concussions).

Games Missed

There was 1 game missed (median) per concussion. The mean burden was 255 games missed annually, ranging from 216 to 286 games per year [games missed imputed for n = 185 (13%)]. Practice concussions resulted in more games missed on average than game concussions (1.6 vs 0.9; $P < 0.01$); median games missed was 1.0 for both. Preseason concussions resulted in more games missed on average than regular season concussions (1.4 vs 1.0; $P = 0.02$; median was 1.0 for both). The distribution of games missed for preseason and regular concussions is shown in Figure 4.

Return to Game Participation

Among players sustaining a preseason or regular season concussion, 41% returned to game play without missing any games, 37% returned after missing one game, and 22% returned after missing ≥2 games (Table 3). Fewer players returned without missing a game in 2019 (n = 72; 39%) and 2020 (n = 51; 30%) compared with prior years (2015-2018: n = 85-105; 41%-45%; Table 3). Tight ends and offensive linemen had the highest proportion of players returning to game play after missing 2 or more games (n = 67; 30% and n = 36; 29%), whereas special teams roster players (kickers, punters, long snappers), wide receivers, and defensive secondary players had higher proportions returning without missing any games (n = 10; 100%, n = 87; 48%, and n = 149; 46%).

In the regular season, 92.9% (52/56) of concussions that occurred among players during a Sunday game and with a Thursday game 4 days later did not participate. This proportion was 97.3% (36/37) from 2017 to 2020, with the single case of a Thursday return in this four-year period because of delayed concussion diagnosis. Among the 52 concussions among players who were withheld from the Thursday game, 65.4% (34/52) returned to competition the following weekend. Among players concussed in a Sunday game, 37.7% (200/531) competed in the next week's Sunday game (if scheduled). This percentage was lower in 2019 (34.2%; 26/76) and 2020 (31.6%; 25/79) than 2015 to 2018 (37.1%-42.4%).

DISCUSSION

National Football League players who sustained concussions during the 2015 to 2020 seasons missed a median of 9 days (mean = 15.0); 59% of in-season concussions caused at least 1 missed game. The estimated league-wide annual concussion burden was 3591 days and 255 games missed, representing missed opportunity for an individual to participate, high medical management by the player-patient and medical staff, and impact to the team caused by absence of players. Continued efforts to reduce concussions, such as updated kickoff rules, restrictions on lowering the helmet, elimination of blindside blocks, and prohibition of poorer performing helmets,[12,23] and to return players safely after sufficient time for full recuperation are of paramount importance.

The RTFP protocol developed and published by the NFL HNS Committee[12] in consultation with the NFLPA's Mackey-White Player Health and Safety Committee follows the concept of a graduated progression using a phased approach presented by the CISG.[16–19] The player starts with limited activity and subsequently progresses to higher levels of exertion, adding strength training and sport-specific activities before introducing light and then full contact. This phased approach has been adopted at all levels of sport. In the NFL, activity progression between protocol phases is supervised by club medical staff and requires that: (1) symptoms are not exacerbated, (2) symptoms return to the athlete's baseline

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TABLE 2. Days Missed Because of Concussions in the NFL, 2015 Through 2020*

	N	Mean (SD)	Median (IQR)	Range	N missing	% Missing	Total Days + Imputed Days by Median†	6-yr Average, Total Days + Iimputed Days by Median†
Overall	1480	15.0 (23.7)	9 (5-14)	0-318	118	8.0%	21 836	3639
Year‡								
2015	284	14.4 (20.9)	9 (5-16)	0-181	37	13.0%	3926	N/A
2016	255	12.7 (18.0)	9 (5-13)	1-155	29	11.4%	3138	N/A
2017	302	17.2 (27.8)	9 (5-14)	0-168	21	7.0%	5029	N/A
2018	223	15.5 (29.3)	8 (5-13)	0-318	13	5.8%	3370	N/A
2019	238	17.4 (27.0)	9 (6-14)	1-224	17	7.1%	4180	N/A
2020	178	12.0 (11.4)	9 (5-16)	3-81	1	0.6%	2180	N/A
Season‡								
Offseason	33	10.4 (8.6)	8.5 (6-12)	0-44	7	21.2%	329.5	55
Preseason	435	17.2 (28.2)	8 (5-14)	0-181	66	15.2%	7064	1177
Regular season	976	14.3 (21.6)	9 (5-15)	0-318	43	4.4%	13 828	2305
Postseason	36	15.5 (30.8)	9 (4-10)	2-155	2	5.6%	545	91
Activity‡								
Game	1181	14.6 (22.7)	9 (5-14)	0-318	80	6.8%	17 028	2838
Practice	297	17.1 (27.4)	8 (5-15)	0-181	38	12.8%	4757	793
Other	2	6.5 (2.1)	6.5 (5-8)	5-8	0	0.0%	13	2
Roster position‡								
Offensive line	261	17.2 (22.6)	10 (6.5-18)	0-154	25	9.6%	4334	722
Running back	117	14.2 (24.6)	9 (5-12)	3-181	10	8.5%	1605	268
Tight end	151	21.3 (37.3)	9.5 (6-17)	0-318	13	8.6%	3242	540
Quarterback	49	8.7 (7.4)	7 (4-10)	2-45	2	4.1%	424	71
Wide receiver	222	15.5 (27.3)	8 (5-13)	1-224	21	9.5%	3368	561
Defensive secondary	366	12.8 (17.5)	8 (4-13)	0-151	22	6.0%	4567	761
Defensive line	133	11.6 (14.1)	8 (5-12)	2-94	9	6.8%	1526	254
Linebackers	169	16.2 (24.8)	9 (6-14)	0-168	14	8.3%	2648	441
Special teams	12	8.6 (15.3)	4 (4-4)	2-52	2	16.7%	94	16
Game day‡§								
Saturday-Tuesday	965	14.5 (22.9)	9 (4-14)	0-318	50	5.2%	13 762	2294
Wednesday-Friday	216	15.2 (21.9)	9 (6-13)	0-144	30	13.9%	3266	544
Practice day‡¶								
Sunday–Wednesday	182	17.0 (27.2)	8 (6-15)	0-181	23	12.6%	2928	488
Thursday–Saturday	115	17.1 (27.8)	7.5 (5-13)	2-160	15	13.0%	1821.5	304

** Players sustaining a concussion in 2020 but had not yet returned to football activity at the time of manuscript submission and did not leave the club with whom they sustained their concussion are assumed to have a return date of February 12, 2021 (n = 8 concussions).*
† Includes days missed during a second bout of missed time following initial RTFP for n = 11 concussions.
‡ Uses imputation by mean/median of each specific category separately for days missed.
§ Includes only concussions sustained during games.
¶ Includes only concussions sustained during practices.

level before initiating contact play, and (3) subsequent neurocognitive and neurologic examinations are at baseline levels before a player is cleared to return to contact play. Unlike some RTFP protocols, a minimum of 24 hours between successive phases is not required. Final RTFP decisions are made by the team physician, which must be confirmed by an independent neurological consultant.[12]

Despite widely accepted consensus statements[6–9,16–19] and sport-specific SRC protocols,[12] limited literature quantifies time lost from concussions in professional American football. Pellman et al [24] first reported data on NFL game-related concussions from 1996 to 2001. Although the median days missed in those 6 years was similar to the 9 days reported here, this examination of NFL concussions more than 10 years later demonstrates an overall more conservative approach to concussion management. Notably, the 1996 to 2001 cohort reported that 57% of concussed players returned to the same game in which they were injured and 97% returned to play within 9 days. In contrast, no player in this 2015 to 2020 cohort returned to play in the same game by protocol mandate, and 59% of players missed at least one game, an indirect but meaningful comparison. Players missed an

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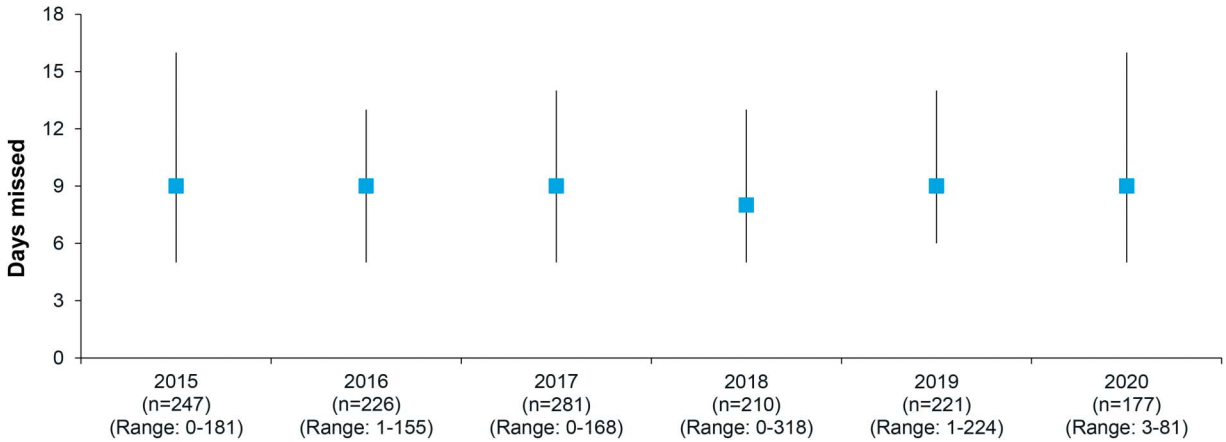

Figure 3. Median and IQR days missed because of concussions in the National Football League, 2015 through 2020. Players sustaining a concussion in 2020 but had not yet returned to football activity at the time of manuscript submission and did not leave the club are assumed to have a return date of February 12, 2021 (n = 8). Players with concussions missing a RTFP were excluded from calculations of median days missed (n = 118).

average of 15 days after game-related concussions. These differences may be attributed to a multitude of factors including adoption of a more cautious approach to concussion management by team medical providers, increased willingness of players to report symptoms due in part to league-wide educational initiatives, safety-focused rules changes, evolution of protocols prohibiting same-day return, increased public scrutiny regarding concussions, and enhanced use of neuropsychological testing.

These data align with a National Collegiate Athletic Association (NCAA) Concussion Study (1999-2001) comparison with NCAA-Department of Defense Concussion Assessment, Research and Education (CARE) Consortium (2014-2017) data[25,26] evaluating clinical management and time to RTFP. Of note, CARE athletes reported symptoms for longer periods of time, had longer symptom-free waiting periods, and longer time to RTFP than the NCAA Study athletes. Compared with the CARE RTFP median of 12.23 days, this NFL study demonstrates a shorter median of 9 days. Similar to the collegiate level, RTFP has increased compared with earlier years, with Casson reporting 1.92 days for NFL players (1996-2001)[27] and NCAA reporting 3.00 days (1999-2001).[28]

The NFL HNS Committee's Concussion Diagnosis and Management Protocol [12] is reviewed annually and updated by NFL and NFLPA committees. Protocol updates occurred throughout the study period based on observed data and evolving clinical evidence, including: a mandatory game-day protocol using the NFL sideline assessment tool,[12] implementation of booth athletic trainer and Unaffiliated Neurotrauma Consultant programs[29,30] requiring players with specific observable signs to be removed and evaluated, and a requirement that concussed players are removed without return to game-day participation. Efforts were also continuously made to standardize the recognition, assessment, and management of SRC, in protocol and in practice.[11,12] Health

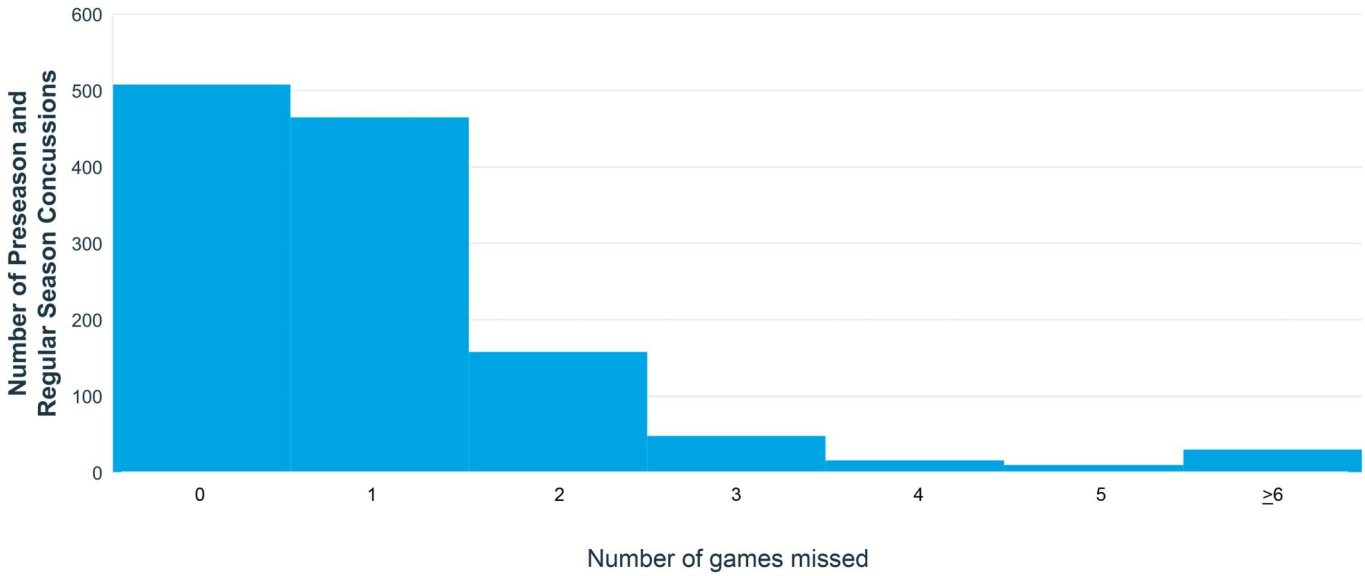

Figure 4. Distribution of games missed because of preseason and regular season concussions in the National Football League, 2015 through 2020. Players with concussions missing an unknown number of games were excluded from this figure (n = 149).

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TABLE 3. Games Missed After Concussions in the NFL, 2015 Through 2020 Preseason and Regular Season Concussions*

	Returned After 0 Games		Returned After 1 Game		Returned After 2 or More Games		Missing Number of Games Missed
	N	Row %†	N	Row %†	N	Row %†	N
Overall	513	40.6%	467	37.0%	282	22.3%	149
Year							
2015	105	42.2%	84	33.7%	60	24.1%	26
2016	99	45.4%	80	36.7%	39	17.9%	25
2017	101	40.7%	97	39.1%	50	20.2%	34
2018	85	45.0%	64	33.9%	40	21.2%	26
2019	72	38.7%	71	38.2%	43	23.1%	38
2020	51	29.7%	71	41.3%	50	29.1%	0
Season							
Preseason	115	33.2%	148	42.8%	83	24.0%	89
Regular season	398	43.4%	319	34.8%	199	21.7%	60
Activity							
Game	458	43.6%	378	36.0%	215	20.5%	96
Practice	54	25.7%	89	42.4%	67	31.9%	52
Other	1	100.0%	0	0.0%	0	0.0%	1
Roster position							
Offensive line	61	27.7%	92	41.8%	67	30.5%	27
Running back	42	42.4%	41	41.4%	16	16.2%	12
Tight end	43	34.7%	45	36.3%	36	29.0%	20
Quarterback	17	39.5%	19	44.2%	7	16.3%	2
Wide receiver	87	48.1%	60	33.1%	34	18.8%	26
Defensive secondary	149	45.7%	107	32.8%	70	21.5%	28
Defensive line	49	41.5%	49	41.5%	20	16.9%	11
Linebackers	55	39.0%	54	38.3%	32	22.7%	21
Special teams	10	100.0%	0	0.0%	0	0.0%	2
Game day‡							
Saturday–Monday	372	42.7%	318	36.5%	181	20.8%	60
Wednesday–Friday	86	47.8%	60	33.3%	34	18.9%	36
Practice day§							
Sunday–Wednesday	26	20.0%	62	47.7%	42	32.3%	30
Thursday–Saturday	28	35.0%	27	33.8%	25	31.3%	22

** Number of games missed was estimated as the number of games the player's team played between the date of the player's removal from participation due to the concussion to the date the player was able to RTFP; for game concussions, the game in which the concussion occurred is not counted regardless of when in the game the player sustained the concussion. The number of games missed is influenced by the timing in the season of the injury.*
† Calculations for row percents exclude those missing return date.
‡ Results include only concussions sustained during games.
§ Results include only concussions sustained during practices.

and safety of NFL players is maximized using these standardized procedures performed by trained medical staff alongside the graduated, progressive return to participation protocol and a comprehensive postinjury neuropsychological assessment battery performed and interpreted by neuropsychologists, medical oversight, and additional medical resources to track progress. The stability of the data over past years and cautious management (not allowing same day return to play) compared with data from a decade ago, demonstrate a careful approach to concussion treatment in professional American Football players reflecting what has occurred at other levels of play.

The NFL RTFP protocol is not meant to be prescriptive. Rather it empowers the player and clinical team to tailor timelines for safe RTFP based on actual observed recovery from the concussive episode. Specifically, the concussed player can clear multiple phases in a single day if he remains asymptomatic as he progresses through the protocol. This progression is defined through self-report by the player, close observation by the medical staff, and detailed neurologic, postural stability, and neurocognitive examinations. Clearance for RTFP is given by the club physician and confirmed by the independent neurological consultant after it is determined the player's concussion symptoms have fully resolved.

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Although the current manuscript does not correlate concussion symptom count and severity with days to RTFP, prior research has found that symptom count and severity at time of concussion is the best predictor of RTFP. Therefore, natural variation in days to RTFP is expected as symptom count and severity likely varied in our study: the <1% of concussions with 0 to 2 days missed in our study is a result of this variation of symptom count and severity.[31]

Time missed per concussion differed by position, with quarterbacks and special teams players missing less time. Special teams players had the third highest total burden (485 days per year) and comprised the third highest proportion of concussions. This difference may be because of variation in magnitude of forces sustained between these groups, which is the basis for the ongoing study.

This study has limitations. Days and games missed are influenced by timing of the injury during the season (late-season concussions have less opportunity to miss games) and personnel decisions (eg, Injured Reserve or players leaving a club). Non-concussion related factors (eg, concurrent injuries or coaching/roster decisions) could affect post-concussion game participation and are not accounted for. Although a return to participation protocol is strongly recommended, treatment (eg, medication, physical therapy) is at the discretion of team medical personnel. In some cases, concussion onset date differs from the diagnosis date, although diagnostic process improvements are in place to reduce delayed diagnoses and reporting.[10,22,28] Days and games missed were imputed for concussions missing return data, which were more complete later in the study as EHR reporting matured. Burden metrics focus on population-wide injury reduction efforts but do not account for individual-level factors (eg, quality of life, pain, medical care). The 2020 season differed from prior seasons because of COVID-19, which may have influenced RTFP. Finally, a concussion may go undetected because of a player not reporting concussion symptoms[32]; however, protocols have been instituted by the NFL to reduce the risk of missed detection or players failing to report symptoms. For example, all players with a stinger or other neck injury are also evaluated for a concussion; analyses have shown that less than 2% of players with game concussions are cleared in the game, but later diagnosed, suggesting that the detection program is highly sensitive.[29,30,33]

CONCLUSIONS

Median time missed after concussion was 9 days, which may be related to emphasis on graduated phase-based concussion management. Approximately one-third of players participate in games the week following a concussion. No concussed players returned to competition on the day of injury. Further work is needed to distinguish characteristics of concussions that result in longer RTFP and to assess the impact of factors such as previous concussion history, symptom type, duration and intensity, and neuropsychological test performance.

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